

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I V.Shiva(192311361) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V-Shiva

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Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 = Exemplary	Level 3 = Proficient	Level 2 = Developing	Level 1 = Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Smart Home Automation system using Intelligent Agents Introduction.

A Intelligent Agent is a system that perceives the environment using sensors and performs action using actuators to achieve specific goals.

- Motion
- Temperature
- Light intensity
- Time of day
- Door/window

The system controls

- Lights
- Air conditions
- Heater
- Door locks.

1. simple Reflex Agent:-

Definition

A simple Reflex Agent acts only on the current perception using predefined IF-Then Rules.

It does not remember previous events

working in smart Home:-

Example Rule :-

If motion detected $\rightarrow$ Turn on lights.

If temperature $> 30^{\circ}\text{C}$ $\rightarrow$ Turn on AC

If doors opens at night $\rightarrow$ Trigger alarm.

2. Model-Based Agent

A model Based Agent maintains an internal model of the environment and remembers previous states. This memory allows the system to make better decision even when sensors information is incomplete unavailable.

- * Maintains an internal memory.
- * Remembers previous states
- * Handles partially observable environments.
- * Makes more accurate.
- * Better than a simple Reflex Agent.

3. Goal Based Agent

A Goal-Based Agent makes decisions by considering a specific goal that need to be achieved. Instead of home, the goals may include maintaining a comfortable room temperature reducing electricity consumption and improve home security.

Utility Based Agent

A Utility Based Agent selects the actions that provides the highest overall benefits by considering multiple factors such as comfort, energy consumption, security and user satisfaction. This enables the system to make a decision that satisfy both the user and the environment.

eg:- Temperature = 27°C

option 1:-

- AC at 18°C
- High comfort
- High electricity consumption

option 2

- fan + AC 25°C
- Good comfort.
- Low electricity consumption.

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A Robot placed inside an unknown maze must search for the exit without having any prior knowledge of the maze structure. This problem can be solved using uninformed search algorithm because the robot has no information about the location of the goal. uninformed cost search [ucs] and Iterative Deepening search [ids]

Example:

Path A $\hat{=}$ cost 5

Path B $\hat{=}$ cost 8

Path - Robot selects path A because it has the lowest cost.

uniform cost search

uniform cost search is an uninformed search algorithm that always expands the nodes with the lowest cumulative path cost. Therefore it guarantees path with the minimum total cost. if all costs are positive.

Iterative Deepening Search

Iterative Deepening Search combines the advantages of Depth first search and Breadth first search. It repeatedly performs depth-limited search by increasing the search depth one level at a time until the goal is found. That is while still guaranteeing that a solution will eventually be found.

Example:

Depth 0 $\rightarrow$ Search Root

Depth 1 $\rightarrow$ search children

Depth 2 $\rightarrow$ search grand children

Continue until Exit is found.

Advantages:

uniform cost search:

- * finds the least cost-path
- * complete algorithm
- * optimal solution
- * These are Advantages of uniform cost search.

Iterative Deepening search:

- * uses very little memory
- * complete algorithm
- * suitable when the goal depth is unknown
- * These are the Advantages of the Iterative Deepening search.

Depth 0 $\rightarrow$ Search Root
Depth 1 $\rightarrow$ Search children
Depth 2 $\rightarrow$ Search grand children
continue until exit is found

Advantages:

uniform cost search:

- * finds the least cost path
- * complete algorithm
- * optimal solution

These are Advantages of uniform cost search.